

Technical Backend Document

Human Resources Module — Payroll Mexico

Product: Índice ERP

Module: Human Resources

Section: Payroll

Jurisdiction: Mexico

Objective: Define calculation logic, data structures, and business rules required for backend implementation of Mexican payroll, including earnings, deductions, bonuses, commissions, adjustments, ISR (income tax), employment subsidy, IMSS, INFONAVIT, and employer obligations.

1. Core Principle

Backend must be the single source of truth for all calculations.

The UI must not perform any fiscal or labor calculations. It should only capture inputs, display values, toggle options, or request recalculations. All official values must be computed by backend.

This applies to:

- ISR (Income Tax)
 - Employment subsidy
 - Employee IMSS
 - Employer IMSS
 - INFONAVIT
 - SAR
 - Occupational risk
 - Childcare IMSS
 - Payroll tax / state tax
 - Total earnings
 - Total deductions
 - Net pay
 - Total employer cost
-

2. Payroll Flow

2.1 Payroll States

State	Description	Editable
Draft	Payroll in preparation	Yes
Processed	Payroll calculated	Limited

State	Description	Editable
Approved	Payroll approved	No
Paid	Payroll paid	No
Cancelled	Payroll cancelled	No

2.2 Flow

1. Payroll is created per period.
2. Backend loads eligible employees.
3. Backend calculates base salary.
4. Backend loads automatic bonuses.
5. User adds bonuses, commissions, or adjustments.
6. User defines statutory applicability.
7. Backend recalculates totals.
8. User reviews simplified view.
9. User reviews detailed view.
10. User processes payroll.

3. Employee Type

Each employee must include:

```
is_statutory_payroll
```

Value	Meaning
true	Included in official fiscal payroll
false	Internal only (no fiscal calculation)

If false, backend must NOT calculate ISR, IMSS, INFONAVIT, or employer obligations.

4. Payroll Formulas

4.1 Earnings

```
period_salary = daily_salary * days_paid

overtime_amount = overtime_hours * overtime_hour_rate

total_variable_pay = bonuses + commissions + positive_adjustments
```

```
total_earnings = period_salary + overtime_amount + total_variable_pay
```

4.2 Deductions

```
total_statutory_deductions = final_isr + employee_imss + infonavit_discount +  
other_statutory
```

```
total_internal_deductions = loans + other_discounts + negative_adjustments
```

```
total_deductions = total_statutory_deductions + total_internal_deductions
```

4.3 Net Pay

```
net_pay = total_earnings - total_deductions
```

4.4 Employer Obligations

```
total_employer_obligations = employer_imss + employer_infonavit + sar +  
payroll_tax + occupational_risk + childcare
```

4.5 Total Payroll Cost

```
total_payroll_cost = total_earnings + total_employer_obligations
```

5. Variable Concepts

Types

Type	Description
BONUS	Bonus
COMMISSION	Commission
ADJUSTMENT	Adjustment

Flags

Field	Description
taxable	Included in ISR

Field	Description
integrates_sbc	Included in IMSS base
apply_to_statutory_payroll	Fiscal applicability
affects_net_pay	Impacts net
source	SYSTEM or MANUAL
active	Enabled/disabled

6. ISR (Income Tax)

Taxable Base

```
isr_taxable_base = sum of all active taxable statutory concepts
```

Calculation

```
excess = base - lower_limit
marginal_tax = excess * rate
isr_before_subsidy = fixed_fee + marginal_tax
```

Final ISR

```
final_isr = max(isr_before_subsidy - employment_subsidy, 0)
```

7. Employment Subsidy

Applies when:

- Employee is statutory
- Jurisdiction is Mexico
- Base is within subsidy table range

8. IMSS

Employee

```
employee_imss = sum(employee components)
```

Employer

```
employer_imss = sum(employer components)
```

9. INFONAVIT

Types

- NONE
- FIXED_AMOUNT
- PERCENTAGE
- VSM

Example

```
infonavit_discount = base * percentage
```

10. SAR

```
sar = sbc * days_paid * 0.02
```

11. State Payroll Tax

```
payroll_tax = tax_base * rate
```

12. Bonuses

Vacation Premium

```
vacation_premium = vacation_days * daily_salary * rate
```

Christmas Bonus

```
christmas_bonus = daily_salary * days * proportion
```

13. Adjustments

```
net_adjustment = positive_adjustments - negative_adjustments
```

14. Rounding

- Use BigDecimal
 - Internal precision: 6 decimals
 - Final values: 2 decimals
-

15. Recalculation Triggers

- Days worked
 - Bonuses
 - Commissions
 - Adjustments
 - Tax flags
 - Salary changes
-

16. Audit

Store:

- User
- Timestamp
- Field changed
- Old value

- New value
-

17. Data Model (Simplified)

payroll_runs

- id
- company_id
- period
- totals

payroll_run_employees

- employee_id
- salary
- earnings
- deductions
- net

payroll_variable_concepts

- type
 - amount
 - flags
-

18. APIs

```
GET /payroll-runs/{id}  
POST /payroll-runs/{id}/recalculate  
PATCH /employees/{id}/attendance  
POST /employees/{id}/variable-concepts
```

19. Validations

- No ISR if not statutory
 - No edits if Paid
 - No deletion of system concepts
 - No missing tax data
-

20. Executive Summary

Payroll must be a backend-driven calculation engine.

Gross = Salary + Overtime + Variable Pay

Tax Base = taxable concepts

ISR = bracket calculation

Net = Gross - Deductions

Cost = Gross + Employer Obligations

Focus on:

- Configurable tax tables
- Strong audit
- Safe recalculation